

than 60 watts. The chosen discrete components assure fixed frequency of 1 Hz (approximately) at pin 3 of timer IC1 (555). Resistor R1 and potmeter VR1 (1-mega-ohm) allow adjustment of the duty cycle. The higher the duty cycle, the higher the output of the heater.

You can connect up to five 10W heating elements in parallel, totaling 50

watts. The consumption of current will be significantly less if fewer coil elements are connected in parallel through toggle switches S2 through S4. Each of these switches has a 6A rating.

Assemble the circuit on a general-purpose PCB. Mount power transistor TIP120 on a thick heat-sink. Isolate the circuit from the heating elements using

only two wire connections. Use wires that can carry more than 6A current. Fix the coil elements below an aluminium or steel rectangular plate which is at least 1mm thick. Do not forget to insulate the heating plate from the elements. Use the car battery for the power supply with a proper current-carrying-capacity wire. ●

WHITE LIGHT FOR REFRIGERATOR

■ UDAY SHENDE

Normally, the refrigerator lights are yellow in colour and go bad very often. If you want a long-lasting white light for your refrigerator, this circuit is especially for you. The circuit is easy to install inside the refrigerator. Also, it consumes very little power compared to the traditional yellow bulb.

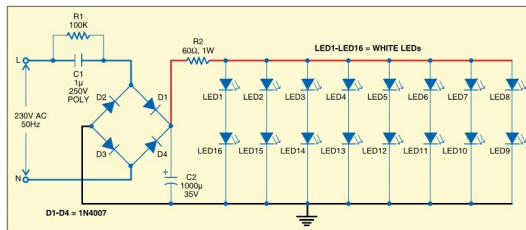
The circuit is transformerless and uses a capacitor for providing the power supply to the white LEDs. Mains input is fed to capacitor C1 and resistor R1, rectified by a bridge rectifier comprising diodes D1 through D4, and filtered by capacitor C2 to provide sufficient voltage to drive the white LEDs. Resistor R2 limits the current flowing through the LEDs. The circuit uses 16 bright



white LEDs in 2x8 parallel-connection configuration.

Assemble the circuit on a general-purpose PCB along with 16 white LEDs. Connect the LEDs in series with

resistor R2. Take out the two wires from the usual bulb connection and connect to the circuit. After connecting the supply wires, wrap electrical tape around the wires properly. Enclose the entire PCB (excluding LEDs) in a plastic box and wrap the box properly with tape to seal it against moisture. ●



SMART HEARING AID

■ DEVRISHI KHANNA AND ROHIT MODI

Normally, hearing aid circuits consume battery power continuously once they are switched on. The circuit given here saves battery power by switching on the sound amplifier section only

when sound is detected. The sensitivity of the detection section and the 'on' time duration of the sound amplifier circuit can be set by the user. Also the circuit uses only a single condenser mic for sound detection and amplification.

As is clear from the above, this hearing aid consists of a condenser

microphone, earphone, and sound detection and amplification sections. The sound detection section employs a quad op-amp IC LM324 (IC1(A)) and a timer NE555 (IC2). The sound signal received at the mic is pre-amplified by transistor BC549 (T1). The voltage

